

Extending the Treatment Window for Intravenous Thrombolysis in Acute Ischemic Stroke

An Updated Systematic Review and Meta-Analysis

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Abstract

Background and Objectives

Intravenous thrombolysis (IVT) is the standard treatment for acute ischemic stroke within 4.5 hours of onset. However, imaging-based selection may extend the treatment window. This systematic review and meta-analysis evaluated the efficacy and safety of IVT administered beyond 4.5 hours after stroke onset or last known well (LKW) in patients selected based on imaging findings.

Methods

A comprehensive search of PubMed, Scopus, and Cochrane Library was performed to identify randomized controlled trials comparing IVT with alteplase or tenecteplase (TNK) administered >4.5 hours after stroke onset/LKW vs standard care. Primary outcomes were 3-month excellent (modified Rankin Scale [mRS] 0–1) functional outcome and symptomatic intracranial hemorrhage (sICH). Secondary outcomes included good (mRS 0–2) functional outcome, recanalization, and 3-month mortality. Pooled odds ratios (ORs) with 95% confidence intervals (CIs) were calculated using random-effects models. Subgroup analysis assessed differences between alteplase and TNK.

Results

Fourteen studies involving 4,944 patients were included. The mean age was 69.8 years, 58.2% were male, the median National Institutes of Health Stroke Scale score was 9, and 12.3% received preplanned endovascular thrombectomy (EVT). A total of 2,492 patients received IVT in an extended time window (4.5–24 hours). Compared with standard care, extended IVT was associated with higher odds of achieving an excellent functional outcome (OR: 1.43 [95% CI 1.25–1.63]), a good functional outcome (OR: 1.25 [95% CI 1.11–1.40]), and recanalization (OR: 3.28 [95% CI 2.09–5.16]). There was no difference in 3-month mortality (OR: 1.21 [95% CI 0.95–1.53]). However, IVT increased the risk of sICH (OR: 2.51 [95% CI 1.47–4.28]). Sensitivity analysis excluding patients who received EVT showed no impact on the outcomes. TNK exhibited similar efficacy to alteplase but showed potentially lower odds of sICH (OR: 1.96, 95% CI 1.06–3.64) compared with alteplase (OR: 5.29, 95% CI 1.80–15.57); however, the subgroup difference was not significant ($p = 0.11$).

Discussion

Among patients selected based on imaging, 4.5–24 hours after stroke onset/LKW, IVT improves outcomes despite an increased risk of sICH. TNK showed similar efficacy to alteplase, with a possible lower risk of sICH; however, direct comparisons in future trials are needed.

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Introduction

Intravenous thrombolysis (IVT) using alteplase has been established as the standard care for acute ischemic stroke (AIS) presented within 4.5 hours of symptom onset.¹ However, strict adherence to this window limits the number of eligible patients, leaving the majority without effective reperfusion therapy. Current clinical recommendations support endovascular thrombectomy (EVT) for individuals with large vessel occlusion (LVO), provided the intervention occurs within 24 hours from the last confirmed time of normal neurologic function.^{1,2} Several factors have driven the need to extend the thrombolysis time window. While EVT has revolutionized ischemic stroke treatment, timely access remains a challenge in many regions. Consequently, IVT remains the only viable treatment for a large portion of the global stroke population.^{3,4}

The 2026 American Heart Association/American Stroke Association guideline gives a new Class 2a recommendation for IVT in patients presenting 4.5–9 hours after onset or with wake-up stroke when advanced imaging shows salvageable tissue. It also offers a Class 2b recommendation for IVT up to 24 hours in select patients with LVO and salvageable tissue who are not candidates for EVT.⁵ Previous studies have produced conflicting evidence regarding the safety and efficacy of IVT beyond the conventional therapeutic window.⁴ Some studies suggest a tailored approach, indicating that certain patients selected using imaging modalities, such as computed tomography (CT) perfusion or MRI with diffusion-weighted imaging (DWI) and perfusion-weighted imaging (PWI) to identify salvageable brain tissue, may still benefit from IVT beyond this therapeutic window.⁶ The WAKE-UP trial demonstrated that MRI-based selection using DWI–fluid-attenuated inversion recovery (FLAIR) mismatch, identifying patients likely to have symptom onset within 4.5 hours, allowed safe and effective treatment with alteplase in patients with wake-up or unknown-onset strokes.⁷ However, this imaging approach identifies only around 62% of cases within the therapeutic window and may overlook patients with potentially viable brain tissue who arrive beyond the 4.5-hour threshold.⁸ A meta-analysis of individual patient data revealed that administering alteplase between 4.5 and 9 hours after stroke onset, or in wake-up stroke patients with imaging-confirmed salvageable tissue, was associated with improved functional recovery. However, this treatment strategy also carried an increased risk of symptomatic intracranial hemorrhage (sICH).²

The latest guidelines from the European Stroke Organization recommend tenecteplase (TNK) at a dose of 0.25 mg/kg as a safe and comparably effective alternative to alteplase for patients with AIS treated within 4.5 hours of symptom onset, based on substantial evidence.⁹ Moreover, the Food and Drug Administration (FDA) has recently approved TNK for use within this therapeutic window.^{5,10} Nevertheless, evidence regarding the effectiveness of TNK in the late therapeutic window remains scarce. In

the OPTION trial, tenecteplase (TNK 0.25 mg/kg) given 4.5–24 hours after onset in patients with non-LVO ischemic stroke and salvageable tissue on perfusion imaging improved excellent functional outcome compared with standard care (43.6% vs 34.2%). It also increased reperfusion and early clinical improvement at 24 hours but was associated with a higher risk of sICH (2.8% vs 0%).¹¹ The lack of direct comparative evidence on the efficacy and safety of alteplase vs TNK complicates decisions on the optimal thrombolytic agent for late-window stroke.^{4,6} The aim of this study was to aggregate data from RCTs to evaluate the efficacy and safety of IVT in patients treated beyond 4.5 hours after symptom onset or last known well (LKW), including wake-up strokes, while also assessing the preferred thrombolytic agent, imaging modality, and therapeutic time window.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

The protocol for this systematic review and meta-analysis was prospectively registered with PROSPERO under the identifier CRD420250655351. The study design and reporting were aligned with the methodological standards outlined in the Cochrane Handbook for Systematic Reviews and Interventions and adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.^{12,13} Because this study constituted a meta-analysis of previously published data, patient consent was not required.

Search Strategy and Data Extraction

A thorough and systematic literature search was conducted to identify relevant studies for inclusion. A team of 5 independent reviewers (S.R., M.A., Y.S., A.M., and A.G.) systematically conducted a literature search across major databases, including PubMed, Scopus, and Cochrane Library. The search strategy was designed using predefined keywords and Boolean operators, including (“tenecteplase” OR “alteplase” OR “thrombolysis”) AND (“stroke”). Any discrepancies identified during the review process were assessed and reconciled through consensus by 2 authors (S.R. and K.H.). The full search strategy is detailed in the supplementary materials (eTable 1). No language, date, or other filters were applied. The search spanned all records from database inception through February 6, 2025, with an additional update conducted on April 28, 2026. To minimize selection bias, the reviewers performed a supplementary hand search by reviewing reference lists of previous systematic reviews, meta-analyses, and conference proceedings to identify additional relevant studies. The study screening and selection process was performed using the Rayyan web-based platform.¹⁴ Data were systematically extracted by 3 authors (M.A., Y.S., and A.M.) using standardized templates, which reported trial names, inclusion and exclusion criteria, sample size, intervention protocols, baseline patient demographics, and predefined efficacy/safety outcomes, with discrepancies resolved by discussion with reviewers (S.R. and K.H.).

Eligibility Criteria

Included studies met these prespecified criteria: (1) RCTs comparing extended IVT with alteplase or TNK with standard care or placebo in selected patients with AIS based on imaging findings for treatment >4.5 hours after LKW or witnessed onset; (2) studies investigating wake-up stroke patients with unknown onset (time from LKW to intervention >4.5 hours); (3) studies reporting at least one predefined outcome of interest. Case reports, review articles, descriptive articles, conference abstracts, and non-RCT original articles were excluded.

Outcomes and Definitions

Efficacy Outcomes

The primary efficacy outcome was a 3-month excellent functional outcome (modified Rankin Scale [mRS] score of 0–1). The predefined secondary efficacy end points included the following, as defined by individual trial protocols: 3-month good functional outcome (mRS score of 0–2); early neurologic improvement (ENI), defined as an National Institutes of Health Stroke Scale (NIHSS) reduction of ≥ 8 points or a score of 0–1 at 24–72 hours^{15–18}; and successful recanalization, assessed as an Arterial Occlusive Lesion (AOL) score of 2–3 at 24 hours, modified Mori grade 3 within 22–36 hours, or Thrombolysis in Myocardial Infarction (TIMI) grade 2–3 at approximately 46 hours.^{3,15,16,19,20}

Safety Outcomes

The primary safety outcome was the incidence of sICH. Additional safety end points included the occurrence of type II parenchymal hemorrhage (type II PH), any form of ICH or systemic bleeding, and mortality at 7 days and 3 months. Comprehensive definitions of all efficacy and safety outcomes are provided in eTable 2.

Quality Assessment

Eligible studies underwent quality assessment and risk-of-bias evaluation using the Cochrane Risk of Bias 2 (RoB 2) tool for RCTs.²¹ Three investigators (M.A., Y.S., and A.M.) independently conducted quality assessments. Discrepancies were resolved through iterative discussion and consensus.

Statistical Analysis

The statistical analysis for this meta-analysis was executed using the R programming language in conjunction with the meta package.²² A random-effects meta-analysis (using the inverse variance method) was undertaken to compute the odds ratio (OR) and determine the 95% CI for pooling the effects derived from individual studies for each respective outcome. Statistical heterogeneity was assessed using the Higgins I^2 test derived from Cochran Q statistics.^{23,24} I^2 values surpassing 50% were deemed indicative of substantial heterogeneity. Sensitivity analyses were conducted using the leave-one-out (LOO) method to evaluate the robustness of the findings. In addition, sensitivity analyses were performed by excluding RCTs in which a substantial proportion of patients underwent preplanned EVT and those that

specifically investigated patients with wake-up stroke. We also performed a subgroup analysis based on the thrombolytic agent used (alteplase vs TNK), treatment time window (4.5 up to 24 hours), and imaging technique used for identifying eligible patients, including PWI, DWI-FLAIR mismatch, PWI-DWI mismatch, and noncontrast CT/MRI. Finally, publication bias was assessed through a visual inspection of funnel plots, supplemented by the Egger regression test for primary outcomes that included more than 10 studies.²⁵

Data Availability

The data extracted and analyzed during this systematic review and meta-analysis are available from the corresponding author on reasonable request.

Results

Study Selection

A total of 5,501 studies were identified. After the removal of duplicates, 4,128 studies remained. After abstract screening, 23 full-text articles were assessed for eligibility, of which 14 RCTs met the inclusion criteria for the final analysis.^{3,7,11,15–20,26–30} The PRISMA flow diagram systematically documents the process of identifying, screening, and selecting eligible studies (Figure 1).

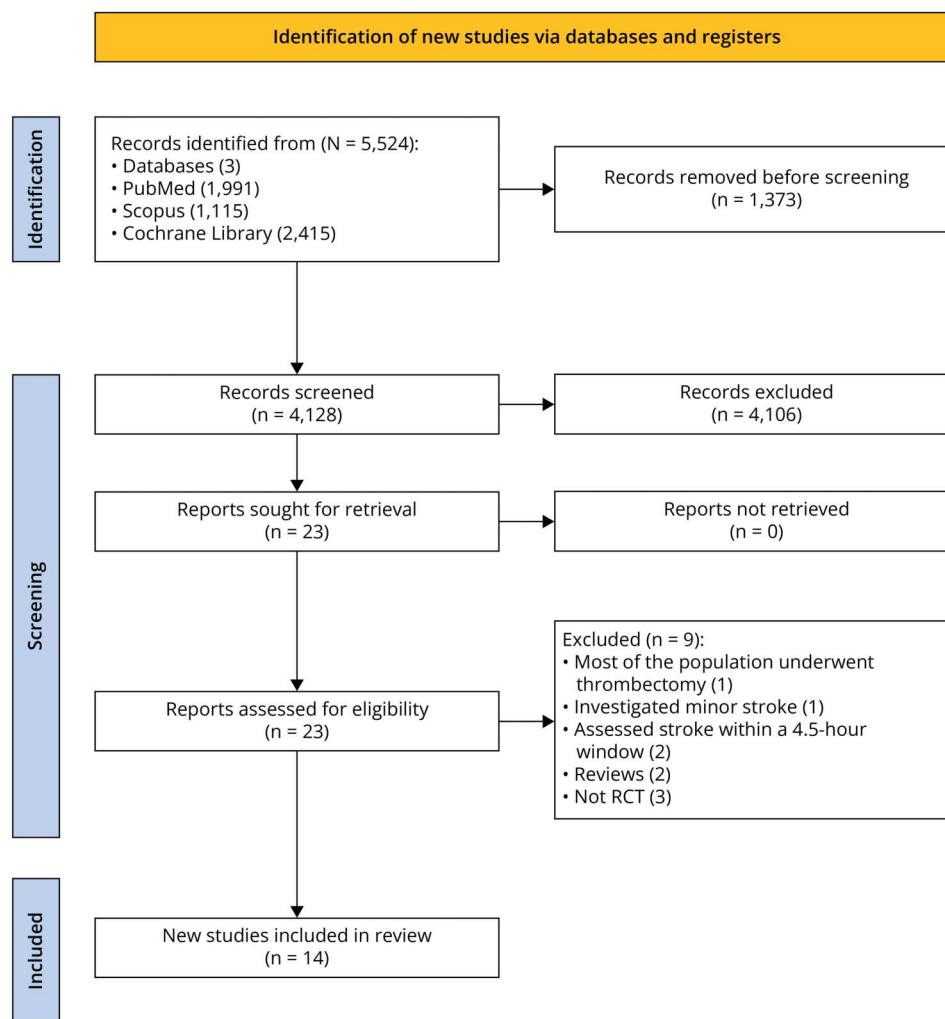
Characteristics of Studies

These 14 RCTs encompassed 4,944 patients, with 2,492 receiving IVT in the extended time window and 2,452 assigned to standard treatment. The average age was 69.8 years, 58.2% were male, and 12.3% received preplanned EVT. The mean time from symptom onset/LKW to treatment was 8.32 hours. The median baseline NIHSS score was 9 in both IVT and standard care groups. The treatment time window varied among studies, ranging from 4.5 to 24 hours. Three studies focused on wake-up stroke patients with unknown onset.^{7,19,28} All 14 RCTs used imaging guidance to inform IVT eligibility. Studies used various imaging techniques: 6 used perfusion imaging (CT-based or MRI-based),^{3,11,15,16,20,26} 3 used DWI/FLAIR mismatch,^{7,17,19} 2 used PWI-DWI mismatch,^{27,30} and 3 used noncontrast CT or MRI.^{18,28,29} The studies were performed across multiple centers and countries. Seven RCTs were multinational,^{7,16,20,27–30} 6 were conducted in China,^{3,11,15,17,18,26} and one was conducted in Japan.¹⁹ The comprehensive characteristics of the incorporated studies are listed in Tables 1 and 2.

Quality Assessment

The quality assessment of the included studies revealed that 5 studies had some concerns^{3,19,27,28,30} and the remaining 9 studies had a low risk of bias.^{7,11,15–18,20,26,29} The primary concerns were related to domains 2 (deviation from intended intervention) and 4 (missing outcome data). A comprehensive summary of the quality assessment is illustrated in eFigure 1.

Figure 1 Prisma Flow Diagram



Primary Efficacy and Safety Outcomes

A random-effects meta-analysis of 13 RCTs^{3,7,11,15–20,26–28,30} evaluating excellent functional outcome (mRS 0–1) at 3 months demonstrated that 42.1% (830/1970) of patients receiving extended IVT achieved mRS 0–1, compared with 34% (662/1994) in the standard care group. Extended IVT was associated with higher odds of achieving mRS 0–1 compared with standard care (OR: 1.43 [95% CI 1.25–1.63], $p < 0.0001$, $I^2 = 0\%$) (Figure 2A).

Twelve of 14 RCTs^{3,7,11,15–20,26–28} provided data on sICH incidence to evaluate the safety of the extended IVT. sICH occurred in 2.9% of extended IVT recipients (57/1932) and 1% of control patients (18/1899). Despite its association with improved functional outcomes, IVT in the extended time window was linked to more than two-fold higher odds of sICH compared with standard care (OR: 2.51 [95% CI 1.47–4.28], $p = 0.0008$, $I^2 = 0\%$)

(Figure 2B). LOO sensitivity analyses confirmed the robustness of these findings, as removing any single study did not significantly alter the overall effect estimates (eFigure 2).

Secondary Outcomes

Based on available data from 14 RCTs^{3,7,11,15–20,26–30} evaluating good functional outcome (mRS 0–2) at 3 months, a more proportion of extended IVT recipients (54.9%, 1,364/2,485) achieved mRS 0–2 compared with 48.9% (1,215/2,449) in the control group. Consistent with the findings for excellent functional outcome, extended IVT was associated with higher odds of achieving mRS 0–2 (OR: 1.25 [95% CI 1.11–1.40], $p = 0.0001$, $I^2 = 0\%$). Extended IVT was associated with greater odds of ENI (OR: 2.26 [95% CI 1.50–3.42]) and recanalization (OR: 3.28 [95% CI 2.09–5.16]). Regarding safety outcomes, extended IVT was associated with higher odds of type II PH (OR: 2.20 [95% CI 1.32–3.64]); however, any ICH (OR: 1.31 [95% CI

Table 1 Baseline Characteristics of Included Studies

Study, year (trial)	Country/ region	Population (IVT/ control)	Age, median (IQR) or mean $\pm$ SD	Male (%)	NIHSS score, median (IQR)	LVO (%)	Baseline mRS inclusion	Time window ^a	EVT (%)	Previous stroke/TIA (%), IVT/ control	HTN (%), IVT/ control	DM (%), IVT/ control	AF (%), IVT/ control
Zhou et al., 2025 (HOPE) ²⁶	China	186/186	72 (62–80)/ 73 (65–80)	54.8/ 59.1	10 (6–15)/ 10 (6–14)	69.3/ 73.7	0–1	4.5–24 h	0/0	17.7/15.6	70.4/ 72.6	26.9/ 25.8	19.4/ 24.2
Cheng et al., 2025 (CHABLIS-T II) ³	China	111/113	49/64.2	71/ 72.1	9 (5–14)/ 9 (6–16)	79.3/ 73.5	0–2	4.5–24 h	27.6/ 1.6	13.5/16.8	64/ 69.9	28.8/ 23	8.1/ 17.7
Albers et al., 2024 (TIMELESS) ²⁰	Multinational	228/230	72 (62–79)/ 72 (62–79)	46.5/ 46.5	12 (8–17)/ 12 (8–18)	57/ 58.3	0–2	4.5–24 h	77.2/ 77.4	NR/NR	NR/NR	N/A/N/ A	N/A/N/ A
Xiong et al., 2024 (TRACE-III) ¹⁵	China	264/252	67 (58–75)/ 68 (59–76)	69.3/ 69.3	11 (7–15)/ 10 (7–14)	NP/ NP	0–1	4.5–24 h	1.5/ 1.98	NR/NR	67/ 71.4	26.1/ 28.2	18.6/ 19
Chen et al., 2024 (EXIT-BT) ¹⁸	China	50/49	63.7/64.8	64.2/ 74	5 (4–7)/5 (4–6)	32.7/ 18.8	0–1	4.5–6 h	0/0	32/36.7	59.2/ 30	30/ 30.6	N/A/N/ A
Ringleb et al., 2019 (ECASS-4) ²⁷	Multinational	61/58	76/79	59/ 59	10 (9/9) (10)	NP/ NP	0–1	4.5–9 h	0/0	NR/NR	N/A/N/ A	N/A/N/ A	N/A/N/ A
Thomalla et al., 2018 (WAKE-UP) ⁷	Multinational	254/246	65.2/65.3	65/ 64.5	6 (4–9)/6 (4–9)	36.5/ 34.1	0–1	$\geq$ 4.5 h (wake- up)	0/0	14.6/12.6	53.1/ 52.6	16.9/ 15.7	11.8/ 11.6
Picanço et al., 2014 (EPITHET) ³⁰	Multinational	38/31	75/73	60/ 60	12 (7–18)/ 11 (8–17)	NR/ NP	0–2	4.5–6 h	0/0	NR/NR	78/63	27/21	46/40
Wang et al., 2023 (ROSE-TNK) ¹⁷	China	40/40	62.68/ 62.8	77.5/ 77.5	7.5 (6–10.7)/ 7 (6–8.7)	42.5/ 0.45	0–1	4.5–24 h	0/0	27.5/30	60/70	0.225/ 0.325	NR/NR
Roaldsen et al., 2023 (TWIST) ²⁸	Multinational	288/290	72.7 $\pm$ 11.3/72.9 $\pm$ 11.6	57/ 58	6 (5–11)/ 6 (5–10)	30/ 37	0–2	$\geq$ 4.5 h (wake- up)	14.5/ 6.2	27/22	64/63	20/19	21/11
Koga et al., 2020 (THAWS) ¹⁹	Japan	70/61	73.2/75.8	64/ 64	7 (4–13)/ 7 (5–12)	NR/ NP	0–1	$\geq$ 4.5 h	0/0	11/23	70/67	20/20	39/34
IST3, 2012 ²⁹	Multinational	507/500	NR/NR	43/ 40.9	NR/NR	NR/ NR	OHS 0–2 (equiv.)	4.5–6 h	NR/ NR	NR/NR	NR/NR	NR/NR	24/22
Ma et al., 2019 (EXTEND) ¹⁶	Multinational	113/112	71/73.2	52.2/ 58.9	12 (8–17)/ 10 (6–16.5)	69/ 72.3	0–1	4.5–9 h	0/0	NR/NR	NR/NR	NR/NR	40.7/ 32.1
Ma et al. 2026 (OPTION) ¹¹	China	282/284	69/67	62.4/ 68.3	7 (5–9)/6 (5–9)	0/0	0–1	4.5–24 h	0/0	NR/NR	74.8/ 74	25.9/ 25.7	12.8/ 13.4

Abbreviations: AF = atrial fibrillation; AIS = acute ischemic stroke; DM = diabetes mellitus; DWI = diffusion-weighted imaging; ENI = early neurologic improvement; EVT = endovascular thrombectomy; FLAIR = fluid-attenuated inversion recovery; HTN = hypertension; IVT = intravenous thrombolysis; LVO = large vessel occlusion; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; sICH = symptomatic intracranial hemorrhage; TNK = tenecteplase.

Data are presented as mean $\pm$ SD, median (interquartile range), number (n), or percentage (%) as indicated.

^aTime from last known well to intervention.

0.91–1.88]), systemic hemorrhage (OR: 1.26 [95% CI 0.59–2.68]), 7-day mortality (OR: 0.94 [95% CI 0.21–4.28]), and 3-month mortality (OR: 1.21 [95% CI 0.95–1.53]) did not differ between extended IVT and standard care. The LOO sensitivity analyses confirmed that none of the secondary

outcomes was susceptible to the exclusion of individual studies. Heterogeneity was low ($I^2 = 0\%$) for most secondary outcomes; however, a substantial heterogeneity was observed for ENI ($I^2 = 50\%$) and recanalization ($I^2 = 60.9\%$) (eFigures 3 and 4). The findings of our study are listed in Table 3.

Table 2 Summary of Trial Intervention

Study, year (trial)	Inclusion criteria	Intervention	LKW-to-IVT (h), median (IQR)	Wake-up stroke (%)	Imaging technique	Occlusion site	LVO (%)	EVT (%)	Ischemic core volume (mL), IVT/control	Treatment window	Neuroimaging criteria
Zhou et al., 2025 (HOPE)²⁶	Potentially salvageable tissue on CTP; no initial plan for EVT	ALT	6.85 (5.47–9.18)	40.9	CT perfusion + CTA	ICA, M1 MCA, proximal/distal M2 MCA, distal MCA beyond M2, ACA, PCA, basilar	69.3	0	12/14	4.5–24 h	Core <70 mL, mismatch ratio ≥1.2, penumbra-core difference ≥10 mL
Cheng et al., 2025 (CHABLIS-II)³	Hypoperfusion (DT > 3 s) to core (rCBF <30) ratio >1.2, difference >10 mL, core <70 mL	TNK	12.33 (8.73–15.9)	N/A	CT perfusion + CTA	Extracranial/intracranial ICA, M1/M2 MCA, ACA	76.3	13.9	6/9	4.5–24 h	Hypoperfusion-core ratio >1.2, difference >10 mL, core <70 mL
Albers et al., 2024 (TIMELESS)²⁰	Core <70 mL, ischemic tissue/core ratio ≥1.8, penumbra ≥15 mL	TNK	12.50 (8.95–16.05)	N/A	CTP or perfusion-diffusion MRI	ICA, M1/M2 MCA	96	77.2	NR	4.5–24 h	Core <70 mL, mismatch ratio ≥1.8, penumbra ≥15 mL
Xiong et al., 2024 (TRACE-III)¹⁵	Core <70 mL, hypoperfused/core ratio ≥1.8, difference ≥15 mL	TNK	12.3 (8.5–16.4)	35.8	CTP or perfusion-diffusion MRI	ICA, M1/M2 MCA	N/A	1.74	16.4/14.9	4.5–24 h	Core <70 mL, mismatch ratio ≥1.8, difference ≥15 mL
Chen et al., 2024 (EXIT-BT)¹⁸	Acute stroke	TNK	5.25 (5.16–5.75)	N/A	Noncontrast CT or MRI	NR	25.2	0	NR	4.5–6 h	None (no advanced selection)
Ringleb et al., 2019 (ECASS-4)²⁷	DWI abnormal but no/negative FLAIR in the same region; clinically acute stroke	ALT	N/A	69	CT or MRI	NR	0	0	NR	4.5–9 h	DWI-FLAIR mismatch (positive DWI, negative FLAIR)
Thomalla et al., 2018 (WAKE-UP)⁷	Wake-up stroke; DWI positive, FLAIR negative/sensitive sequence	ALT	10.3	93.8	MRI (DWI, FLAIR, hemorrhage-sensitive, TOF-MRA)	Intracranial ICA, MCA main stem/branch	35	0	NR	Wake-up	DWI-FLAIR mismatch
Picanço et al., 2014 (EPITHET)³⁰	No perfusion-diffusion MRI mismatch eligibility required	ALT	5.33	NR	MRI	NR	NR	NR	NR	4.5–6 h	None (no mismatch required)
Wang et al., 2023 (ROSE-TNK)¹⁷	No hemorrhagic stroke on CT; DWI-FLAIR mismatch	TNK	12	58.7	CT scan	MCA, ACA, PCA	43.7	0	0.32/0.40	4.5–24 h	DWI-FLAIR mismatch
Roaldsen et al., 2023 (TWIST)²⁸	Wake-up stroke; no ICH, no structural lesions, ischemic lesion ≤1/3 MCA	TNK	10.86	100	Noncontrast CT or MRI	ICA, M1/M2 MCA	26.3	10.38	NR	Wake-up	No advanced mismatch; exclusion of large infarct/ICH
Koga et al., 2020 (THAWS)¹⁹	Wake-up stroke; DWI/FLAIR mismatch, negative FLAIR	TNK	10.2	71	MRI	ICA, M1/M2 MCA, PCA, basilar	N/A	0	3.5/1.9	Wake-up	DWI-FLAIR mismatch
IST3, 2012²⁹	No ICH or structural brain lesions	ALT	NR	NR	Noncontrast CT or MRI	NR	NR	NR	NR	4.5–6 h	None (basic exclusion of ICH/structural lesions)
Ma et al., 2019 (EXTEND)¹⁶	Perfusion lesion-ischemic core mismatch ratio ≥1.2, difference >10 mL, core <70 mL	ALT	5.5	65	CT or MRI perfusion	ICA, MCA M1/M2/M3–4, ACA, PCA, other	39.1	0	4.6/2.3	4.5–9 h	Mismatch ratio ≥1.2, difference >10 mL, core <70 mL

Continued

Table 2 Summary of Trial Intervention (continued)

Study, year (trial)	Inclusion criteria	Intervention	LKW-to-IVT (h), median (IQR)	Wake-up stroke (%)	Imaging technique	Occlusion site	LVO (%)	EVT (%)	Ischemic core volume (mL), IVT/control	Treatment window	Neuroimaging criteria
Ma et al. 2026 (OPTION) ¹¹	Salvageable tissue on penumbral imaging, perfusion-core mismatch ratio ≥ 1.2 , difference >10 mL, core <50 mL	TNK	12	32	CT perfusion	MCA M1/M2/M3-4, ACA, PCA, other	0	0	0 (0-3.7)/1 (0-4.2)	4.5-24 h	Perfusion-core mismatch ratio ≥ 1.2 , difference >10 mL, core <50 mL

Abbreviations: ACA = anterior cerebral artery; ALT = alteplase; CTA = CT angiography; CTP = CT perfusion; DT = delay time; EVT = endovascular thrombectomy; ICA = internal carotid artery; LVO = large vessel occlusion (proximal); MCA = middle cerebral artery; N/A = not applicable; NR = not reported; PCA = posterior cerebral artery; rCBF = relative cerebral blood flow; TNK = tenecteplase; TOF-MRA = time-of-flight magnetic resonance angiography.

Subgroup Analysis

Prespecified subgroup analysis examined main outcomes to assess differences in efficacy and safety between alteplase and TNK in the extended treatment window. Extended IVT was associated with improved functional outcomes at 3 months across subgroups. No differences emerged between alteplase and TNK ($p = 0.49$ for mRS 0-1; $p = 0.58$ for mRS 0-2). Alteplase was associated with higher odds of sICH (OR 5.29; 95% CI 1.80-15.57), whereas TNK showed lower odds (OR 1.96; 95% CI 1.06-3.64). However, the subgroup difference was not statistically significant for sICH across agents ($p = 0.11$). For recanalization, the pooled odds ratio was 3.29 ([95% CI 1.63-6.64]) in the TNK subgroup with high heterogeneity ($I^2 = 80.9\%$) and 3.28 ([95% CI 1.96-5.48]) in the alteplase subgroup with no heterogeneity ($I^2 = 0\%$). The test for subgroup differences showed no difference between the agents ($p = 0.99$). These findings suggest comparable efficacy of TNK and alteplase in achieving recanalization based on available data (eFigure 5).

Subgroup analyses were performed to evaluate the impact of various imaging modalities and treatment time windows on functional outcomes measured at 3 months. PWI (CT-based or MRI-based) emerged as the sole modality linked to significant enhancements in clinical outcomes. Patients selected using PWI exhibited elevated odds of attaining an excellent functional outcome (mRS 0-1; OR: 1.46 [95% CI 1.23-1.74]). This association indicates that PWI provides more effective identification of viable brain tissue amenable to salvage. Other modalities, such as DWI/FLAIR, combined PWI-DWI, and noncontrast CT or MRI, lacked associations with outcome improvements (Table 4). The risk of sICH was consistently elevated in the PWI subgroup (OR: 2.77 [95% CI 1.29-5.95]), but not in the other imaging modalities. This finding points to an elevated bleeding risk among perfusion-selected patients, albeit with limited events contributing to uncertainty. However, subgroup comparisons for imaging modalities yielded no difference in outcomes ($p = 0.95$ for mRS 0-1; $p = 0.87$ for mRS 0-2; $p = 0.98$ for

sICH). Heterogeneity in imaging protocols likely contributed to these results (eFigure 6).

Analyses stratified by treatment window detected no variations in neurologic improvement rates and sICH incidence across strata ($p = 0.94$ for mRS 0-1; $p = 0.99$ for mRS 0-2; $p = 0.82$ for sICH). IVT delivered within an extended window of up to 24 hours after symptom onset demonstrated an association with improved excellent functional outcome (mRS 0-1; OR: 1.46 [95% CI 1.22-1.75]). Such evidence supports the application of IVT in later periods for appropriately selected cases. An elevated sICH risk appeared in the extended window up to 24 hours (OR: 2.22 [95% CI 1.11-4.45]) (eFigure 7). Table 4 presents a detailed summary of subgroup analyses.

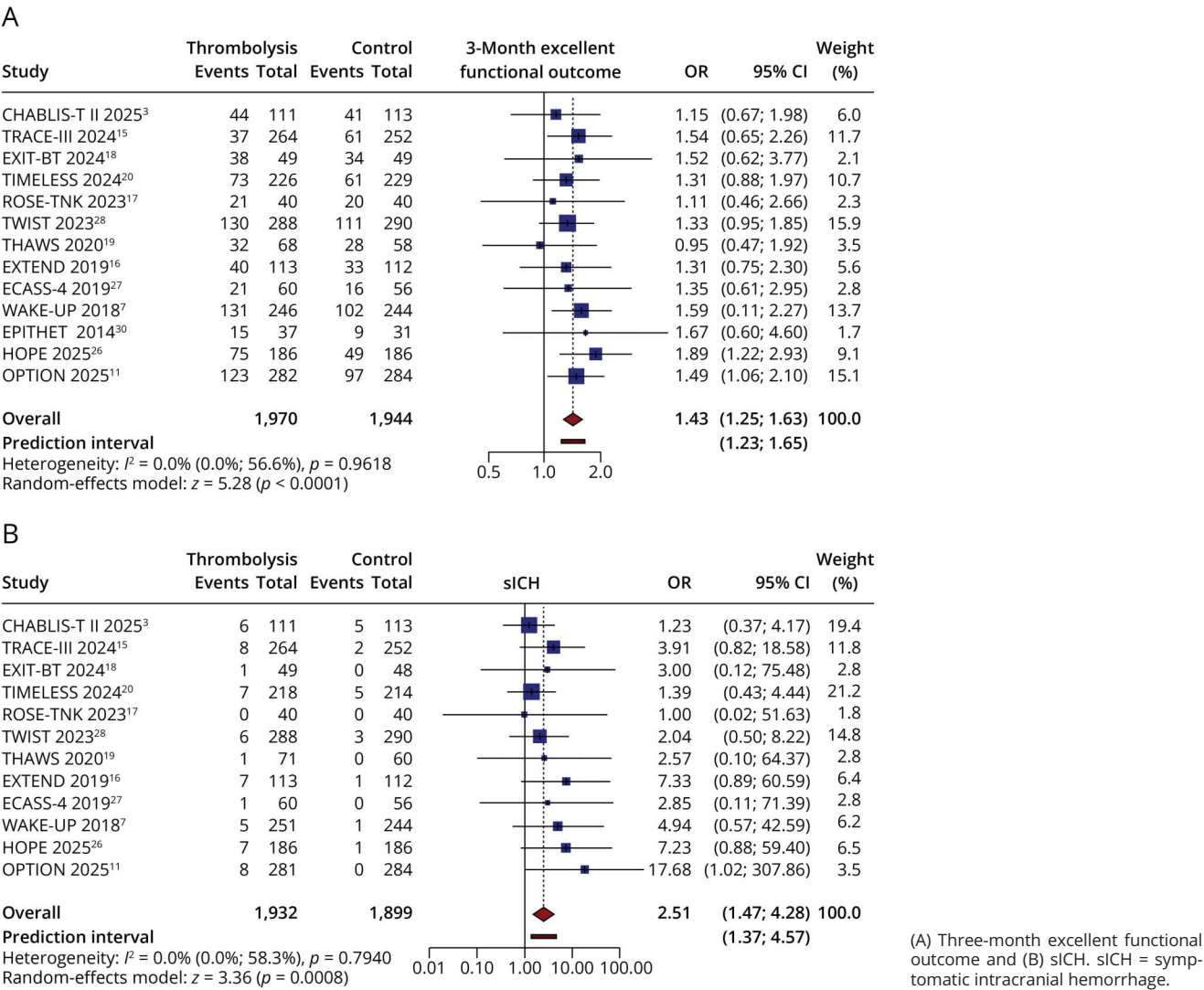
Sensitivity Analysis Assessing the Influence of EVT on the Clinical Outcomes of Extended IVT

Among the included RCTs, only 2 had a substantial proportion of participants who underwent preplanned EVT (54.9% in CHABLIS-T II and 77.3% in TIMELESS). To assess the potential effect of EVT on the pooled estimates, we performed sensitivity analyses by excluding these 2 trials from key efficacy and safety outcomes. These analyses indicated that none of the clinical outcomes were altered by excluding these trials, including excellent and good functional outcomes (mRS 0-1, OR: 1.46 [95% CI 1.27-1.69]; mRS 0-2, OR: 1.30 [95% CI 1.15-1.47]) and mortality (OR: 1.24 [95% CI 0.95-1.63]). However, the odds of sICH increased from 2.51 [95% CI 1.47-4.28] to 3.90 [95% CI 1.95-7.82] (eFigure 8).

Sensitivity Analyses Excluding Wake-Up Stroke Trials

Notably, 3 RCTs specifically investigated extended IVT in wake-up stroke patients using imaging to confirm salvageable penumbra. Sensitivity analyses excluding these RCTs showed no substantial effect on the observed pooled estimates for excellent and good functional outcomes (mRS 0-1, OR: 1.45

Figure 2 Forest Plots for Primary Efficacy and Safety Outcomes



[95% CI 1.24–1.71]; mRS 0–2, OR: 1.26 [95% CI 1.11–1.44]), sICH (OR: 2.54 [95% CI 1.33–4.85]), or mortality (OR: 1.16 [95% CI 0.89–1.51]) (eFigure 9).

Publication Bias

No evidence of publication bias was detected based on the Egger test and funnel plot analysis for excellent functional outcome, good functional outcome, and sICH (Egger test: $p = 0.31$ for mRS 0–1, $p = 0.63$ for mRS 0–2, $p = 0.08$ for sICH) (eFigure 10).

Discussion

The systematic review and meta-analysis demonstrated that extended IVT significantly enhanced excellent and good functional outcomes, early neurologic improvement, and recanalization rates vs standard care in patients selected through advanced imaging beyond 4.5 hours after LKW or witnessed stroke onset.

Extended IVT elevated sICH risk. Mortality rates remained comparable between IVT and control groups in the 4.5–24-hour window. Independent of subsequent EVT, extended IVT is associated with favorable functional outcomes. Patients with wake-up stroke selected by imaging criteria may derive benefit from IVT. Exclusion of wake-up stroke trials exerted minimal impact on pooled outcomes for extended-window IVT. Consistency in clinical outcomes appeared across time strata (6, 9, 24 hours) and imaging modalities. Perfusion imaging showed potential for patient selection and was associated with improved functional outcomes (OR: 1.46 [95% CI 1.23–1.74]). TNK demonstrated efficacy comparable to alteplase and was associated with lower odds of sICH (TNK: OR 1.96, 95% CI 1.06–3.64 vs alteplase: OR 5.29, 95% CI 1.80–15.57); however, the between-subgroup difference was not statistically significant ($p = 0.11$).

Our findings align with previous meta-analyses supporting IVT’s benefit beyond 4.5 hours. For excellent functional

Table 3 Overview of Analyses for Clinical Outcomes

Outcome	Number of studies	Odds ratio (95% CI)	p Value	I ² ; p for Cochran Q
Primary outcomes				
Efficacy outcome				
Excellent functional (mRS 0–1) outcome at 3 mo	13	1.43; (1.25–1.63)	<0.0001	0; 0.96
Safety outcome				
sICH	12	2.51; (1.47–4.28)	0.0008	0; 0.79
Secondary outcomes				
Efficacy outcomes				
Good functional (mRS 0–2) outcome at 3 mo	14	1.25; (1.11–1.40)	<0.0001	0; 0.70
Recanalization	5	3.28; (2.09–5.16)	<0.0001	60.9; 0.03
Early neurologic improvement	7	2.26; (1.50–3.42)	<0.0001	50; 0.06
Safety outcomes				
Any ICH	5	1.31; (0.91–1.88)	0.149	0; 0.41
Type II PH	7	2.20; (1.32–3.64)	0.002	0; 0.68
Systemic hemorrhage	8	1.26; (0.59–2.68)	0.553	0; 0.87
3-mo mortality	11	1.21; (0.95–1.53)	0.116	0; 0.84
7-d mortality	2	0.94; (0.21–4.28)	0.936	0; 0.97

Abbreviations: mRS = modified Rankin Scale; sICH = symptomatic intracranial hemorrhage; Type II PH = type II parenchymal hemorrhage.

outcome (mRS 0–1), our results concord with those of a recent data synthesis⁶ and agent-specific studies.^{31–36} However, good functional outcomes showed mixed evidence in previous literature: 2 TNK studies and one alteplase study reported no benefit over control,^{6,35,36} whereas others favored alteplase^{31,32,34} or TNK.³⁷ This discrepancy may reflect differences in trial populations or dosing (e.g., TNK 0.25 mg/kg vs 0.4 mg/kg in some studies). A detailed summary of findings from similar meta-analyses is provided in Table 5. The improvements in ENI and recanalization rates in our study echo the findings of 2 previous pooled analyses,^{6,34} which also identified these benefits over control. Despite some discrepancies in agent-specific efficacy and comparative studies, our findings supported IVT as a valuable option for delayed stroke presentations.

The increased sICH risk with extended IVT is consistent with previous studies,^{6,31–34,40} although TNK-specific analyses suggested a safer profile.^{6,35–37} Knapen et al.³⁸ observed superior outcomes with standard-window IVT (<4.5 hours) vs extended IVT, indicating that earlier treatment remains preferable when feasible. Nonetheless, our analysis suggests that IVT may retain efficacy and safety in delayed presentation, reinforcing its role when EVT is unavailable.

Alteplase, a recombinant tissue plasminogen activator, has been the long-standing standard.⁴¹ However, its short half-

life (4–6 minutes) necessitates an hour-long infusion (0.9 mg/kg), complicating delivery.⁴¹ By contrast, TNK, a modified variant of alteplase, has a longer half-life (20–24 minutes), allowing for single-bolus administration (0.25 mg/kg) with greater fibrin specificity and resistance to plasminogen activator inhibitor-1, potentially reducing bleeding risk.^{42,43} While alteplase has extensive clinical evidence supporting its use, TNK offers logistical advantages that may improve treatment efficiency. Moreover, studies suggest that TNK may be a cost-effective alternative.^{44,45} This meta-analysis indicates comparable efficacy between TNK and alteplase for thrombolytic therapy. TNK demonstrated similar efficacy to alteplase and was associated with lower odds of sICH (OR 1.96, 95% CI 1.06–3.64) relative to alteplase (OR 5.29, 95% CI 1.80–15.57), although these comparisons should be interpreted with caution, given the exploratory nature of the analysis (Table 4). Consistently, 2 previous studies.^{6,37} reported that TNK's sICH risk was more similar to that of the control group than that of alteplase. Despite the increased sICH risk with extended IVT, mortality remains unaffected, possibly because imaging-based patient selection identifies those with small cores and good collaterals, which leads to fewer fatal hemorrhages, better reperfusion, and more manageable complications. This aligns with previous meta-analyses reporting higher sICH rates without increased mortality.^{6,37} These findings suggest the need for further robust head-to-head trials

Table 4 Overview of Subgroup Analysis for Primary Outcomes

Outcome	3-mo excellent functional outcome			3-mo good functional outcome			sICH		
	No. of studies	Odds ratio (95% CI)	I^2 ; p for Cochran Q	No. of studies	Odds ratio (95% CI)	I^2 ; p for Cochran Q	No. of studies	Odds ratio (95% CI)	I^2 ; p for Cochran Q
Thrombolytic agent									
Alteplase	6	1.52; (1.22–1.89)	0; 0.68	7	1.29; (1.10–1.52)	0; 0.90	6	5.29; (1.80–15.57)	0; 0.87
TNK	7	1.38; (1.17–1.63)	0; 0.96	7	1.20; (1.00–1.45)	17.9; 0.29	6	1.96; (1.06–3.64)	0; 0.66
$p_{\text{difference}} = 0.49$			$p_{\text{difference}} = 0.57$			$p_{\text{difference}} = 0.11$			
Treatment time window									
Up to 24 h	6	1.46; (1.22–1.75)	0; 0.73	6	1.26; (1.04–1.54)	28; 0.22	6	2.22; (1.11–4.45)	7.7; 0.36
Up to 9 h	2	1.32; (0.84–2.09)	0; 0.96	2	1.26; (0.82–1.93)	0; 0.79	2	5.52; (0.94–32.27)	0; 0.63
Up to 6 h	2	1.59; (0.81–3.12)	0; 0.90	3	1.23; (0.97–1.55)	0; 0.83	1	3.00; (0.12–75.48)	NA
Wake-up stroke (>4.5 h)	3	1.38; (1.10–1.73)	0; 0.43	3	1.21; (0.92; 1.59)	12.2; 0.32	3	2.64; (0.88–7.93)	0; 0.80
$p_{\text{difference}} = 0.94$			$p_{\text{difference}} = 0.99$			$p_{\text{difference}} = 0.82$			
Imaging technique/modality									
PWI	6	1.46; (1.23; 1.74)	0; 0.76	6	1.28; (1.07–1.52)	28.3; 0.22	6	2.77; (1.29–5.95)	21.5; 0.27
DWI-FLAIR mismatch	3	1.34; (0.94–1.91)	0; 0.39	3	1.35; (0.99–1.84)	0; 0.50	3	3.18; (0.62–16.24)	0; 0.78
PWI-DWI mismatch	2	1.46; (0.78–2.72)	0; 0.95	2	1.16; (0.66–2.07)	0; 0.99	1	2.85; (0.11–71.39)	NA
Noncontrast CT or MRI	2	1.35; (0.99–1.84)	0; 0.78	3	1.18; (0.97–1.43)	0; 0.68	2	2.16; (0.60–7.79)	0; 0.83
$p_{\text{difference}} = 0.95$			$p_{\text{difference}} = 0.87$			$p_{\text{difference}} = 0.96$			

Abbreviations: CT = computed tomography; DWI = diffusion-weighted imaging; FLAIR = fluid-attenuated inversion recovery; MRI = magnetic resonance imaging; mRS = modified Rankin Scale; PWI = perfusion-weighted imaging; sICH = symptomatic intracranial hemorrhage; Type II PH = type II parenchymal hemorrhage.

comparing TNK and alteplase in the extended treatment window to elucidate potential differences.

The individual patient data meta-analysis of 3 major trials demonstrated that perfusion-guided administration of alteplase in the 4.5–9-hour window or for wake-up strokes significantly improved functional outcomes when used without subsequent EVT, despite an increased risk of symptomatic bleeding.³⁹ By contrast, later studies of TNK given before EVT in extended windows showed improved early reperfusion but failed to demonstrate functional benefits, with potential increased risks in certain stroke subtypes.^{3,20} Sensitivity analyses excluding TIMELESS and CHABLIS-T II (with EVT rates of 77.3% and 54.9%, respectively) did not alter the improvements in functional outcomes or mortality but increased the odds of sICH from 2.51 (95% CI 1.47–4.28) to 3.90 (95% CI 1.95–7.82).

These findings suggest that the excess hemorrhagic risk of extended IVT persists and may be more pronounced in settings with lower EVT utilization.

Our analyses detected no differences across treatment time windows. In the 24-hour window, individual trials varied. CHABLIS-T II indicated a promising trend with uncertainty from inconsistent results.³ TRACE-III delivered strong evidence of benefit.¹⁵ ROSE-TNK yielded neutral outcomes in a small cohort.¹⁷ For the 9-hour window, both EXTEND and ECASS-4 demonstrated positive outcomes.^{16,27} Within the 6-hour window, EXIT-BT used reduced-dose alteplase (0.6 mg/kg vs standard 0.9 mg/kg) and reported improved functional outcomes.¹⁸ EPITHET exhibited modest improvement.^{30,46} IST-3 confirmed limited benefit.²⁹ HOPE and OPTION demonstrated functional benefits of extended

Table 5 Findings From Previous Meta-Analyses That Assessed IVT in the Extended Time Window

Author	Design of included studies in meta-analysis	Year	Total number of patients	Comparison	Findings							
					Efficacy				Safety			
					3-mo excellent functional outcomes	3-mo good functional outcomes	Recanalization	Early neurologic improvement	sICH	Any ICH	PH2	3-mo mortality
Aladawi et al. ³⁷	Conference abstract (5 RCTs)	2025	1,197	TNK vs control	NR	Higher in TNK	NR	NR	Similar	NR	NR	NR
Kelani et al. ³⁵	Conference abstract (4 RCTs)	2025	1,629	TNK vs control	Higher in TNK	Similar	NR	NR	Similar	NR	NR	Similar
Günkan et al. ⁶	8 RCTs	2025	1,742	TNK vs control	Higher in TNK	Higher in TNK	Higher in IVT	Higher in IVT	Similar	Higher in IVT	Higher in IVT	Similar
				ALT vs control	Higher in ALT	Similar	Higher in IVT	Higher in IVT	Higher in IVT	Higher in IVT	Higher in IVT	Similar
Al-Janabi et al. ⁴	8 RCTs	2024	2,221	TNK vs control	Higher in TNK	Similar	NR	NR	Similar	NR	NR	Similar
				ALT vs control	Higher in ALT	Higher in ALT	NR	NR	Higher in ALT	NR	NR	Similar
Knapen et al. ³⁸	12 observational studies	2024	1,589	Extended IVT vs standard IVT	NR	Higher in standard IVT	Similar	NR	Higher in extended IVT	NR	NR	Higher in extended IVT
Palaodimou et al. ³⁶	3 RCTs	2024	556	TNK vs control	Higher in TNK	Similar	NR	NR	Similar	Similar	NR	Similar
Jia et al. ³¹	4 RCTs	2021	849	ALT vs control	Higher in ALT	Higher in ALT	NR	NR	Higher in ALT	NR	NR	Similar
Thomalla et al. ³²	4 RCTs	2020	843	ALT vs control	Higher in ALT	Higher in ALT	NR	NR	Higher in ALT	NR	Similar	Higher in ALT
Chen et al. ³³	12 RCTs	2019	3,402	ALT vs control	Higher in ALT	NR	NR	NR	Higher in ALT	NR	NR	Higher in ALT
Tsivgoulis et al. ³⁴	4 RCTs	2019	859	ALT vs control	Higher in ALT	Higher in ALT	Higher in ALT	Higher in ALT	Higher in ALT	NR	NR	Similar
Campbell et al. ³⁹	3 RCTs	2019	414	ALT vs control	Higher in ALT	NR	NR	NR	Higher in ALT	NR	NR	Similar
Emberson et al. ⁴⁰	9 RCTs	2014	6,756	ALT vs control	Similar	NR	NR	NR	Higher in ALT	NR	Higher in ALT	Similar

Abbreviations: ALT = alteplase; ICH = intracranial hemorrhage; IVT = intravenous thrombolysis; NR = not reported; Type II PH = type II parenchymal hemorrhage; RCT = randomized controlled trial; TNK = tenecteplase.

IVT up to 24 hours after stroke onset.^{11,26} These trials collectively support thrombolysis efficacy in extended windows. Among wake-up stroke trials assessing extended IVT, TWIST showed no advantage,²⁸ THAWS indicated no benefits,¹⁹ and WAKE-UP revealed significant improvement.⁷ A patient-level meta-analysis provided robust evidence of IVT benefit in wake-up stroke guided by advanced perfusion imaging.³⁹ In sensitivity analyses excluding wake-up stroke trials,^{7,19,28} efficacy and safety outcomes remained unchanged (eFigure 9).

Subgroup analyses stratified by imaging techniques indicated minimal impact of modality choice on IVT efficacy or safety in the extended time frame. Perfusion imaging showed potential for patient selection and was associated with improved functional outcomes (mRS 0–1; OR: 1.46 [95% CI 1.23–1.74]) (eFigure 6). It has been proposed that imaging criteria affect IVT efficacy.⁶ Perfusion imaging, used in EXTEND¹⁶ and TRACE-III,¹⁵ was associated with stronger functional improvements. DWI-FLAIR mismatch, used in WAKE-UP⁷ and THAWS,¹⁹ exhibited weaker associations. Choice of imaging modality/technique had minimal impact on the outcomes (p for subgroup difference >0.05) (eFigure 6). Heterogeneity in eligibility criteria and thrombolytic agent used across trials likely introduced outcome variability across imaging modalities/techniques. This supports application in resource-limited settings lacking advanced imaging. A study compared multimodal CT and MRI in AIS of unknown onset. CT-based selection achieved shorter treatment times than MRI. No significant differences emerged in bleeding risk or long-term favorable outcomes between modalities.⁴⁷ RCTs collectively endorse neuroimaging to detect perfusion lesion-ischemic core mismatch as a valid selection method for thrombolysis candidates with AIS.^{14,25,47} Neuroimaging criteria details are listed in Table 2.

Our systematic review and meta-analysis builds on and expands the findings of the previous studies. A recent pooled analysis⁶ excluded trials that allowed preplanned EVT in their participants, whereas, by including these trials, we investigated extended IVT in a more pragmatic setting. Bridging therapy with IVT before EVT has been investigated in previous studies and has yielded mixed findings.^{48,49} An individual participant data meta-analysis⁴⁹ showed that IVT combined with EVT improved outcomes in individuals with anterior circulation large-vessel occlusion compared with EVT alone, with significant benefits observed only when the interval from symptom onset to IVT administration was short.

Further strengths include a broader range of efficacy and safety end points and a comprehensive evaluation of both alteplase and TNK, similar to 2 recent meta-analyses^{4,6} while contrasting with previous extended-window evidence syntheses that often focused on a single agent.^{31,32,35–37} We also conducted a detailed subgroup analysis of various imaging techniques and modalities to explore their potential impact on the clinical outcomes of extended IVT.

A limitation of our study is that we evaluated only thrombolytics currently used in clinical practice and did not include others, such as desmoteplase, as investigated in other studies.⁵⁰ In addition, many patients in the included RCTs were enrolled based on the time of LKW rather than stroke onset. Consequently, patients with wake-up stroke or unknown onset time may have experienced a stroke within 4.5 hours of LKW. Although we conducted sensitivity analyses excluding wake-up stroke trials, isolating patients with confirmed stroke onset beyond 4.5 hours was not feasible because of insufficient data. Moreover, the variability observed in trial designs, particularly differences in patient selection, may affect pooled estimates. However, low heterogeneity was observed among analyses, reinforcing the comparability of clinical outcomes across studies. Our analysis did not account for potential differences in stroke etiology or location (e.g., anterior vs posterior circulation) because most trials did not stratify these factors. Finally, focusing on 3-month outcomes misses longer term effects on recovery or mortality, which could be relevant for assessing IVT's sustained benefit in delayed treatment.^{4,6,36} These limitations highlight areas for future research to build on our findings.

IVT administered 4.5–24 hours after stroke onset or LKW improves functional outcomes and recanalization despite the higher sICH risk. There was no difference in clinical outcomes across various treatment windows. The imaging modality/technique used had minimal influence on outcomes, although perfusion imaging showed potential for better patient selection and was associated with improved functional outcomes. TNK demonstrated efficacy comparable to alteplase and may be associated with a lower risk of sICH compared with alteplase, although definitive conclusions require confirmation through direct head-to-head trials.

Author Contributions

S. Rahmati: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. M. Alinejadfar: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. Y. Zarinfar: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Moradi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Ghabousian: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Moradi: drafting/revision of the manuscript for content, including medical writing for content. K. Najafi: drafting/revision of the manuscript for content, including medical writing for content. F. Iskander: drafting/revision of the manuscript for content, including medical writing for content. A. Latib: drafting/revision of the manuscript for content, including medical writing for content. T. Kuno: drafting/revision of the manuscript for content, including medical writing for content.

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